

Getting UDT1.1

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Summary: The below is a way to obtain in-the-limit UDT1.1 behavior. It can seem hacky or ad hoc, but I think it's a good start.

Main idea: For further and further nodes in the decision tree, we condition $\mathbb{P}$ on larger and larger policies (instead of single actions). For example, for the first 10 rounds we just condition on single actions as usual, but then, from round 10 to round 20, we condition on all possible conjunctions of actions taken in those rounds (across all possible worlds). And then from 20 to 120, etc. Given some assumptions about

1. not getting infinitely screwed by early actions¹ (catastrophe-free),
2. and not being "so unlucky" that time and time again the important entanglements land between the different unconnected policy regions, despite these boundaries having density 0,

we get all desired behavior in the limit.

This also works in the realistic case: It might seem like the above assumes that all share the same representations of decision trees. But that's not the case: if two agents think they inhabit the same decision tree, but they have represented it differently (for example, changing the order of branches or actions without breaking any structure), the protocol above will nonetheless work, since the utilities conditional on (isomorphic) policies will turn out the same.

It's true that, in the realistic situation where each agent has a probability distribution over decision trees, it'll be harder for them to easily coordinate (they need good knowledge about their decision tree), and they'll constantly be make expected value trade-offs between possible trees they inhabit. But that's of course a fundamental complication of not knowing your environment's source code, and if two agents entertain the same hypothesis, it will not matter how they represent it.

Might this be fragile on implementation? Agents will also be uncertain about the round number. And depending on the round number, the agent can or cannot coordinate with this or that agent (because we are optimizing over smaller or larger policies). But I think this uncertainty is in

¹Although this no longer makes sense if we use time-penalized finite utility for our infinite decision tree, right? Any action can just lock away finitely much utility. Maybe we should say "it doesn't lock away infinitely much utility BEFORE time-penalization", but that feels unnatural, and is not translatable to the more natural case in which, instead of taking an infinite tree and time-penalizing it, we just assume that each infinite branch sums to finite utility (and maybe also something like "these finite values are bounded"). Maybe the conditional result to be proven here should just read "for decision trees in which an arbitrarily high fraction of the total (finite) utility is arbitrarily far away from the earliest nodes, we get arbitrarily few coordination failures".

no way more worrisome than the above general empirical uncertainty (it's just one more piece of it). And additionally, this worry in particular should decrease for later rounds, since there boundaries between policy regions are rarer, and so it's less likely to mess up "which policy size should I be optimizing now".

Can we nonetheless do better? $\mathbb{P}$ is an infinite distribution on all logical sentences (despite its information seed being the finite $\mathbb{Q}_k$). Instead of increasing the finite policy horizon as above, what if we just take the infinite policy horizon from the start? We can indeed "condition" $\mathbb{P}$ on an infinite conjunction (of action points), by taking the limit of the conditional probability (about, say, total U obtained) for larger finite initial segments of it. A priori this might not converge... but given our assumption that each branch has a finite amount of utility, PLUS these amounts are bounded, I think it does always converge! (Since arbitrarily far-off nodes must yield an arbitrarily small change in U .) Also, taking an arbitrary limit is not computable, but given the "information seed" is finite, probably some decidable proof system will prove the value of the limit (by just taking into account all of the finite information).

We might worry that this is just "letting the earliest prior decide all actions, without updating in any way, and so this doesn't learn". Again, I think we can edit the prior to hard-code something like "I don't yet know what the best action here will be, but (maybe because I believe strongly that there are no dangerous correlations) it will be at least as updateful as looking at $\mathbb{Q}_n$, and implementing this same procedure of infinite policy selection for the infinite tail of that part of the tree".